

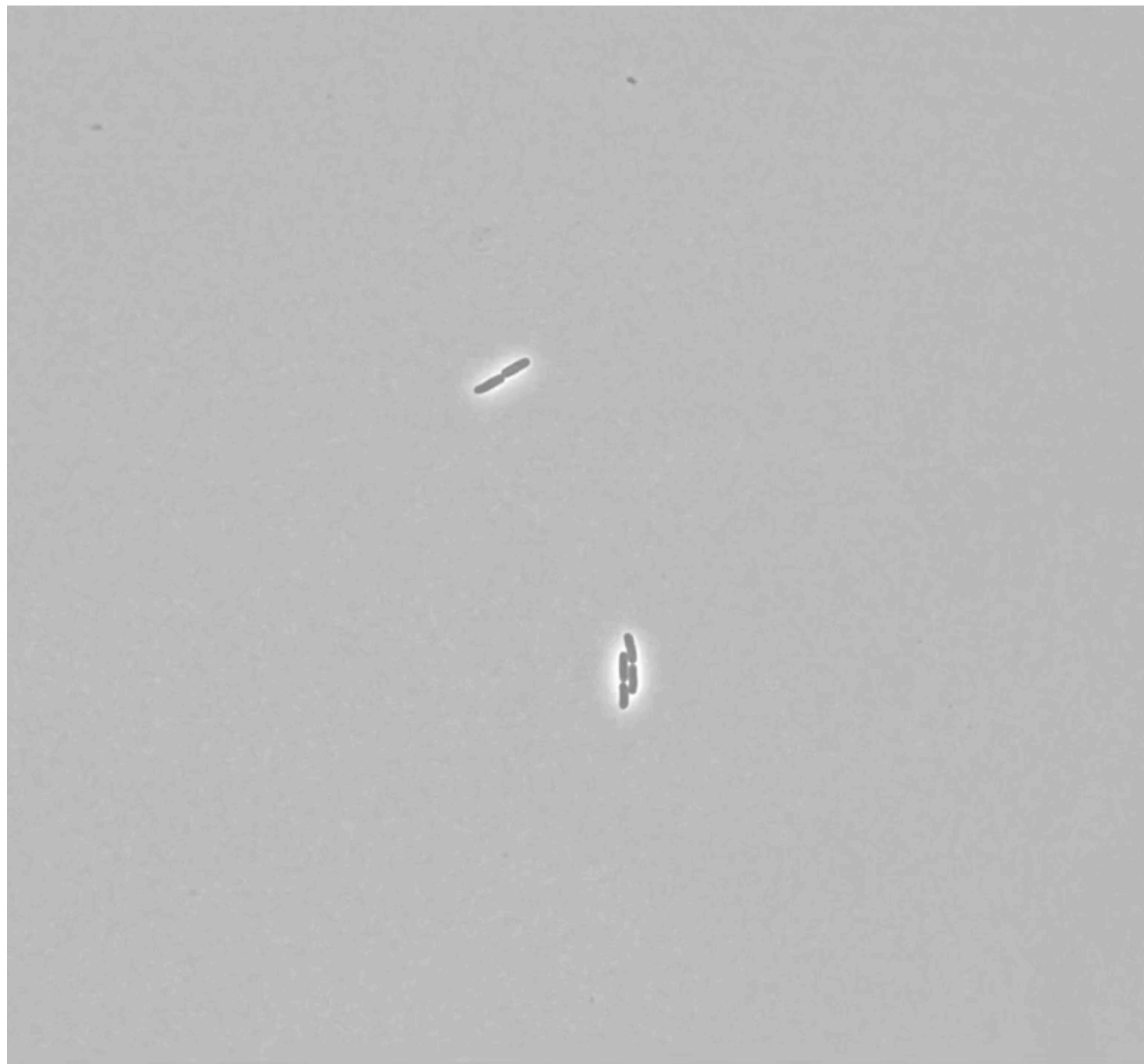
L3

Mutations in expanding populations

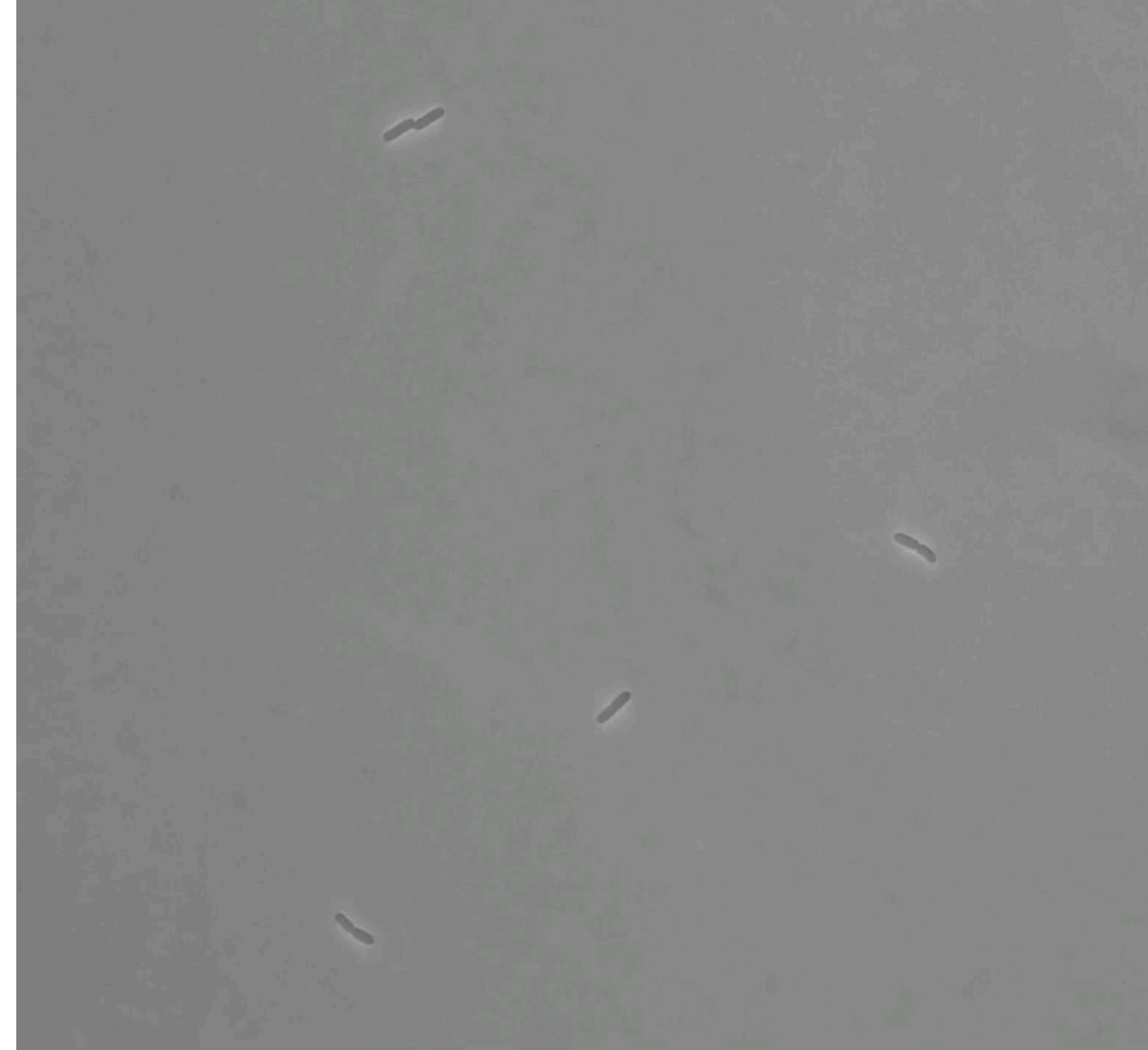
Jacopo Grilli - jgrilli@ictp.it - <http://jacopogrilli.github.io>

Modulate growth rates by killing cells

slow growth (bacteriostatic)

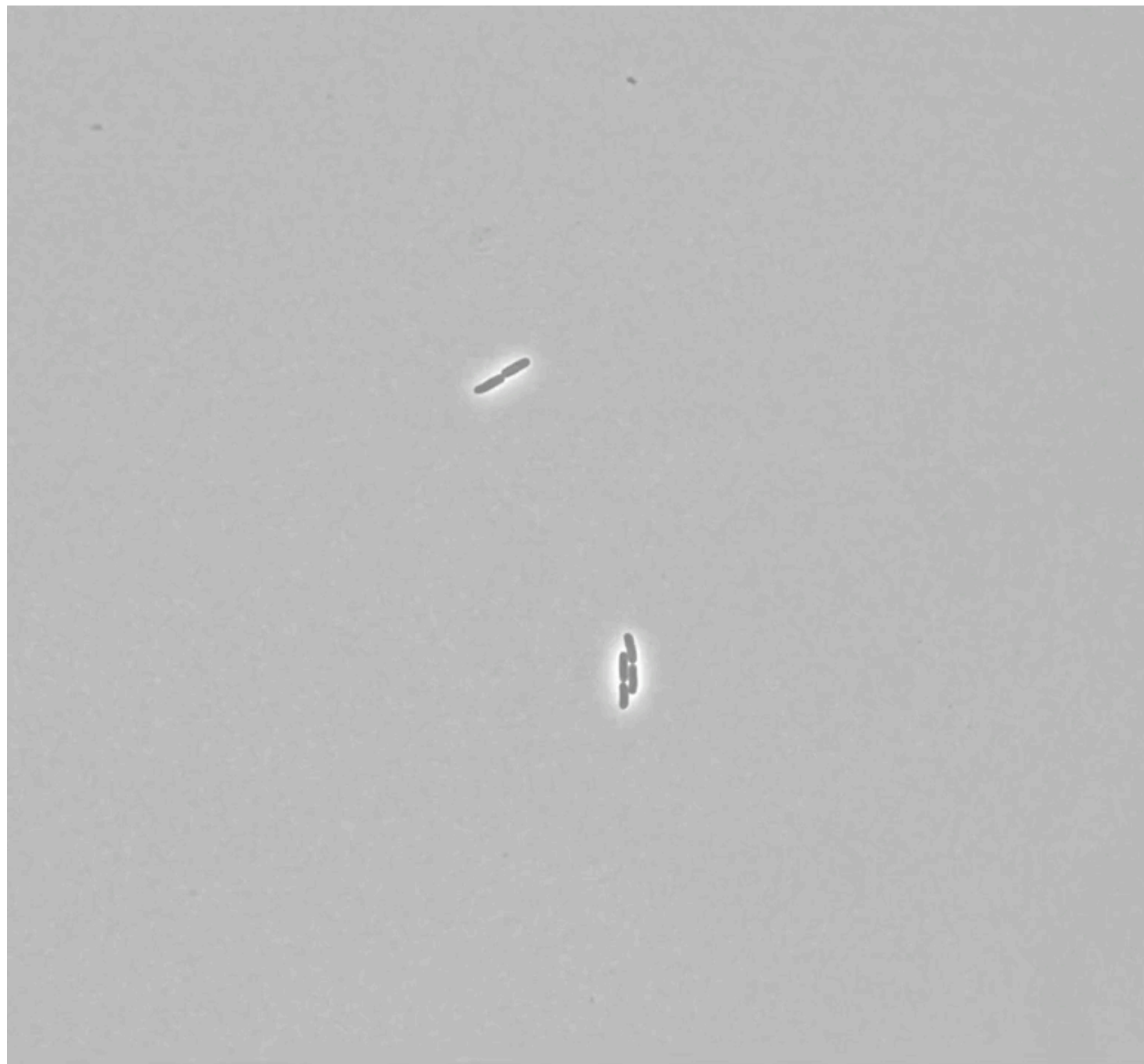


kill cells (bactericidal)

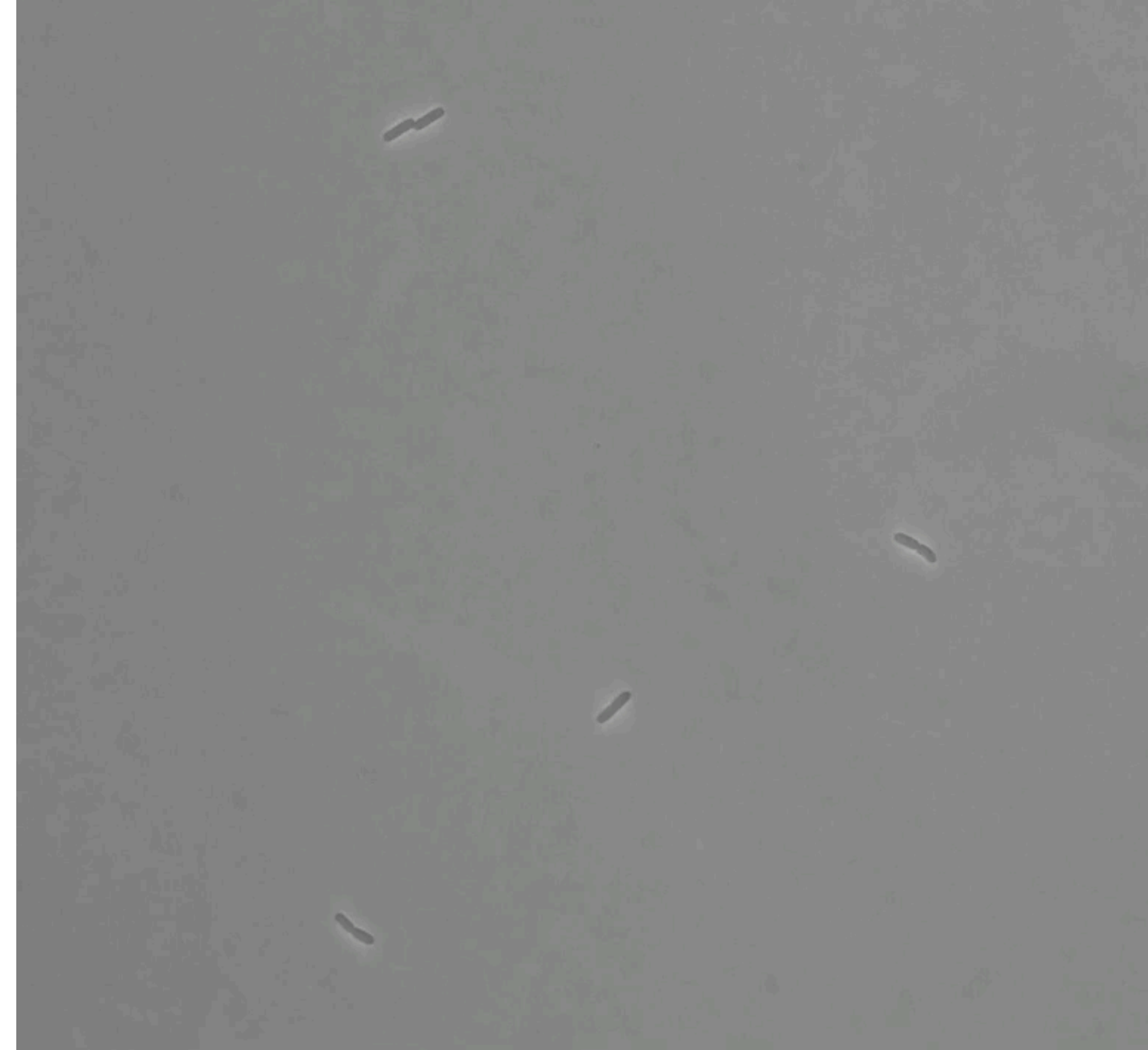


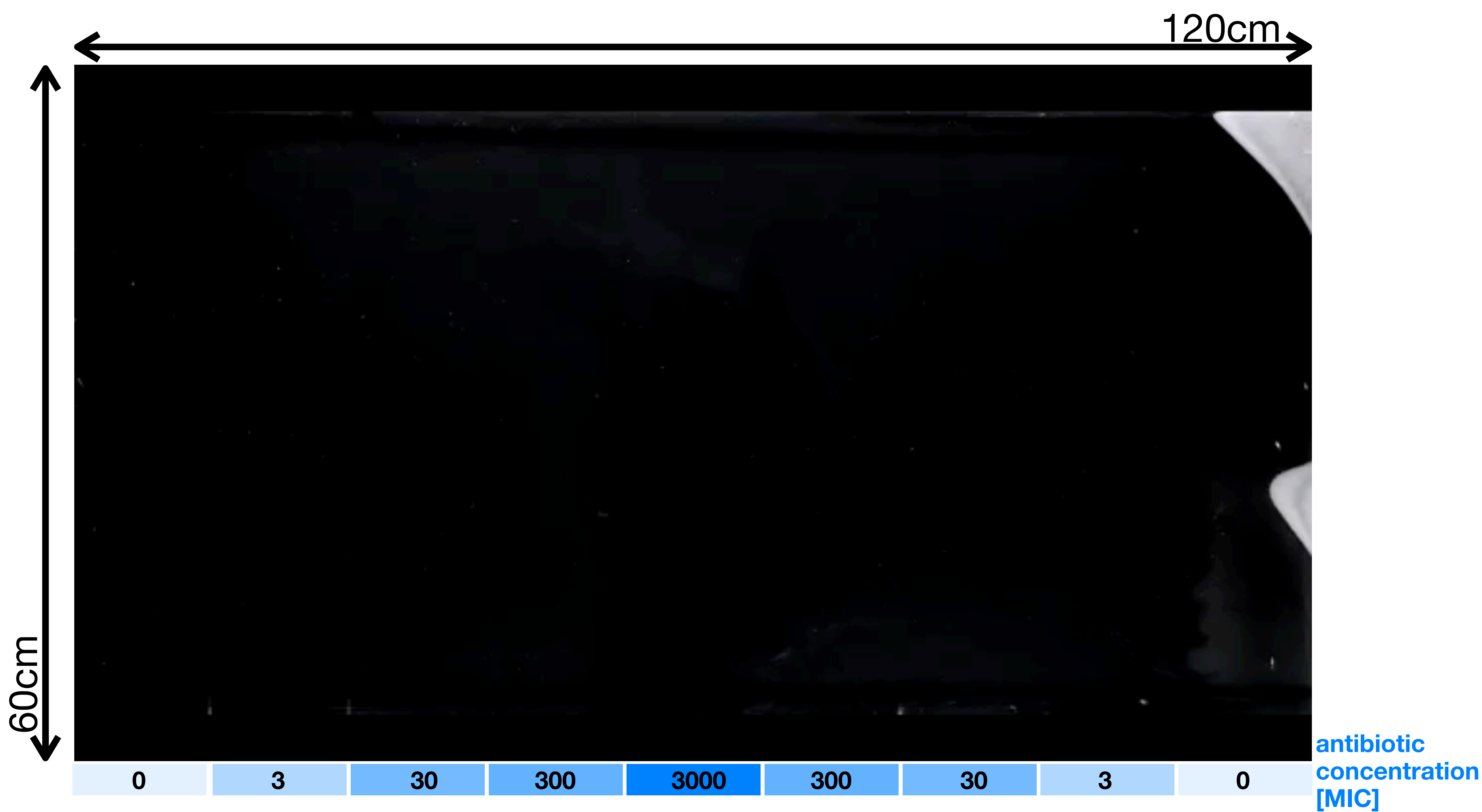
Modulate growth rates by killing cells

slow growth (bacteriostatic)



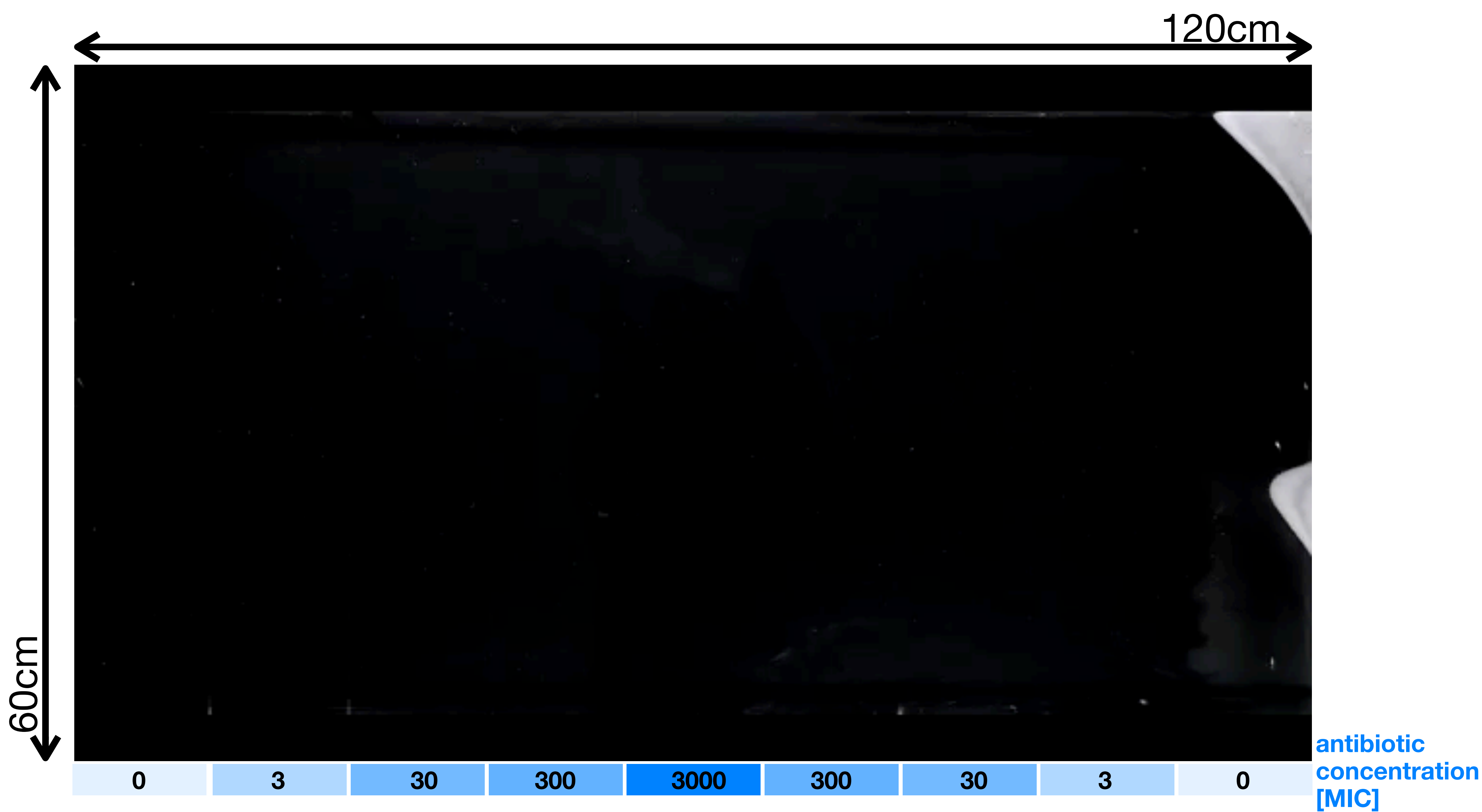
kill cells (bactericidal)





total duration: 11 days

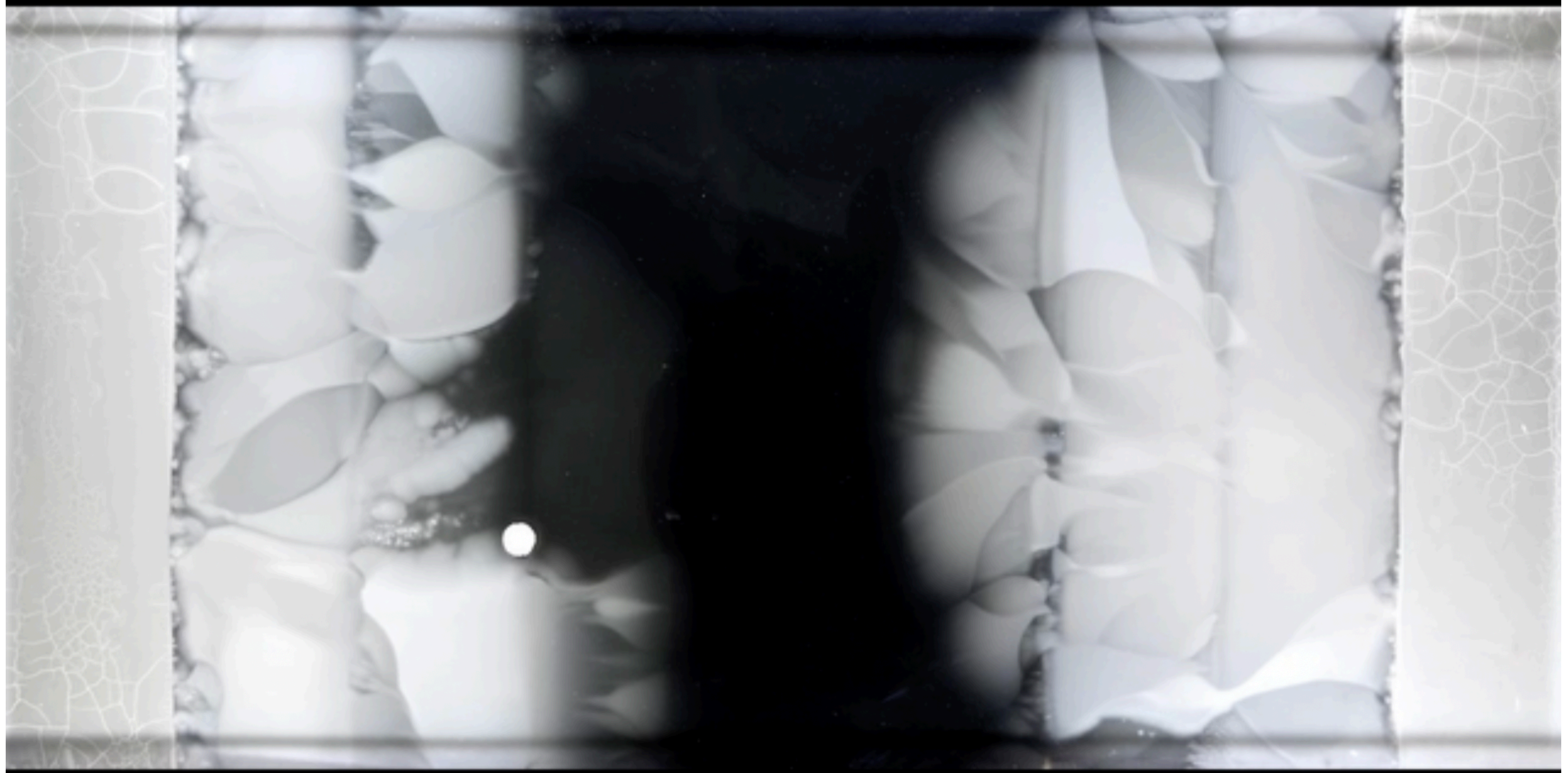
[10.1126/science.aag0822](https://doi.org/10.1126/science.aag0822)



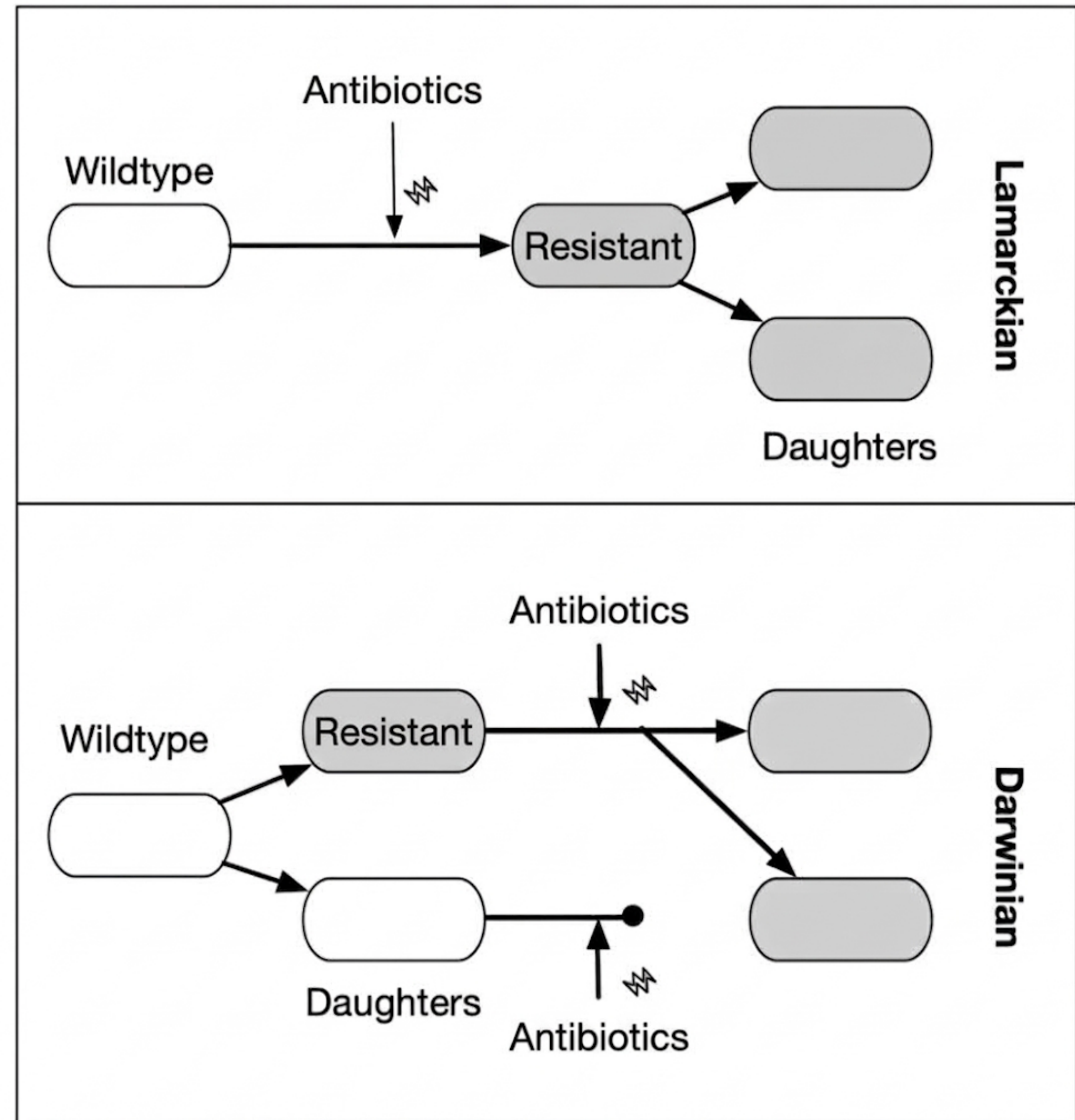
total duration: 11 days

[10.1126/science.aag0822](https://doi.org/10.1126/science.aag0822)

What's going on?



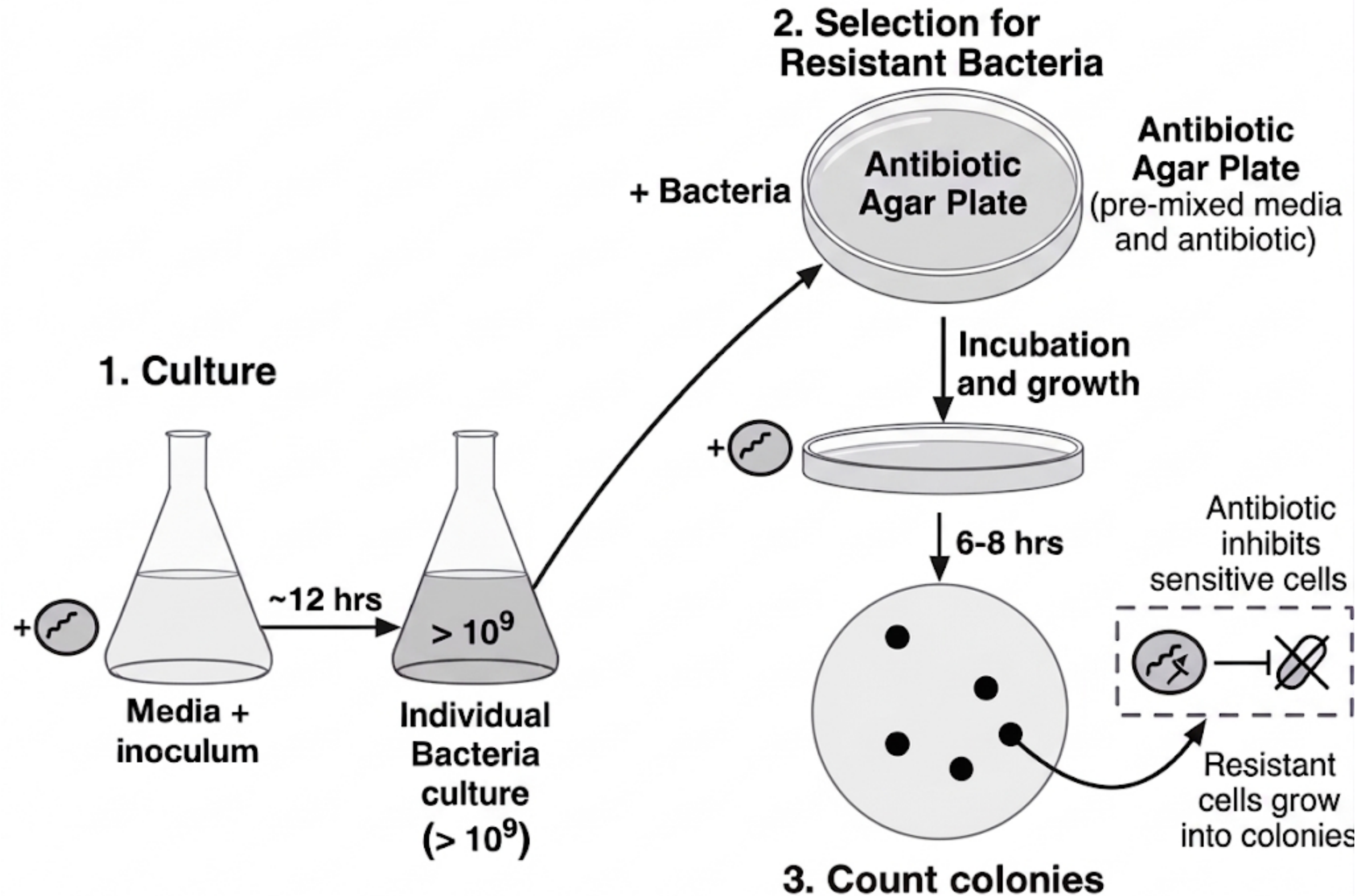
Two alternatives



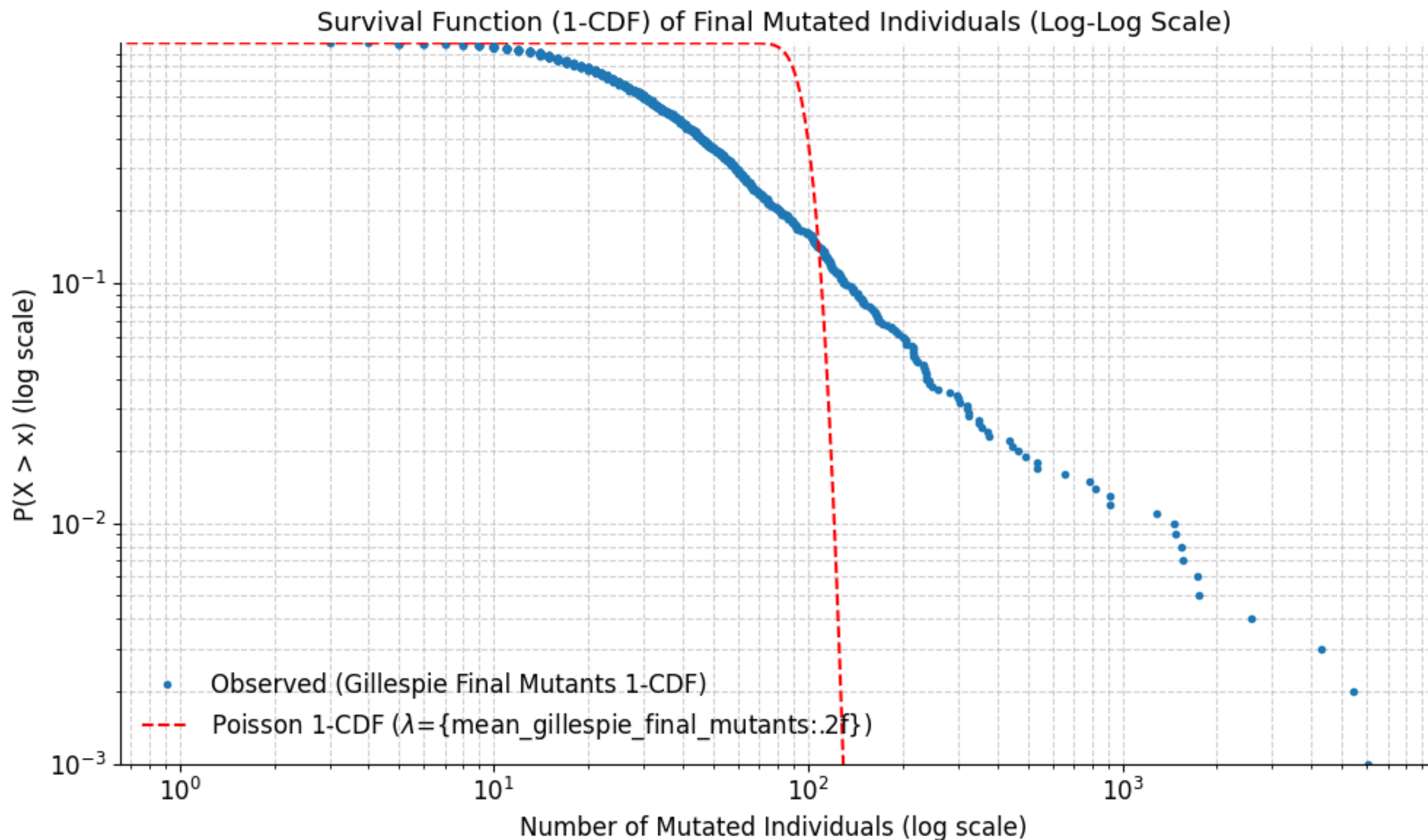


Salvador Luria and Max Delbrück

Luria-Delbrück experiment



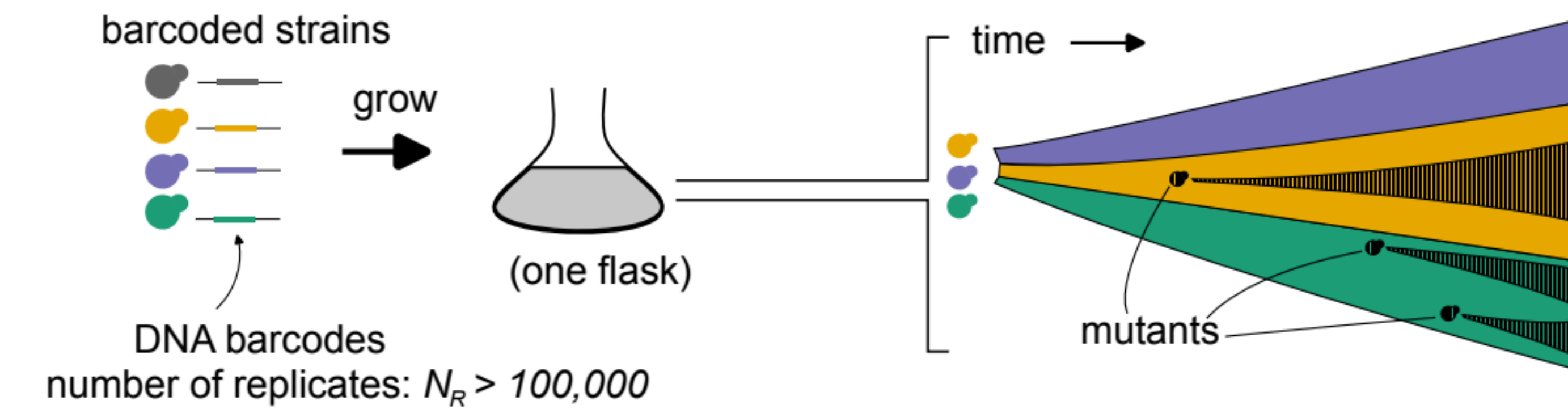
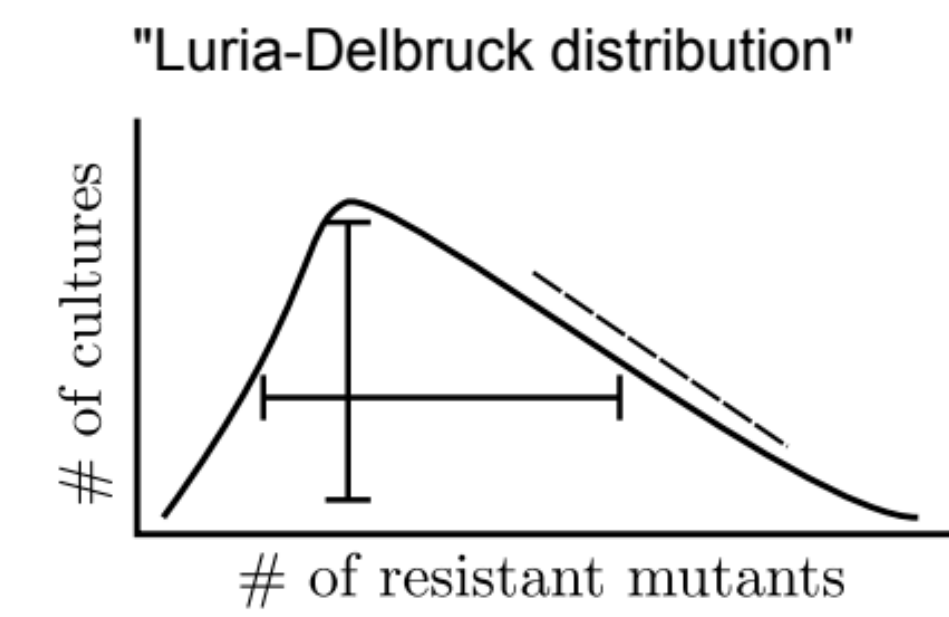
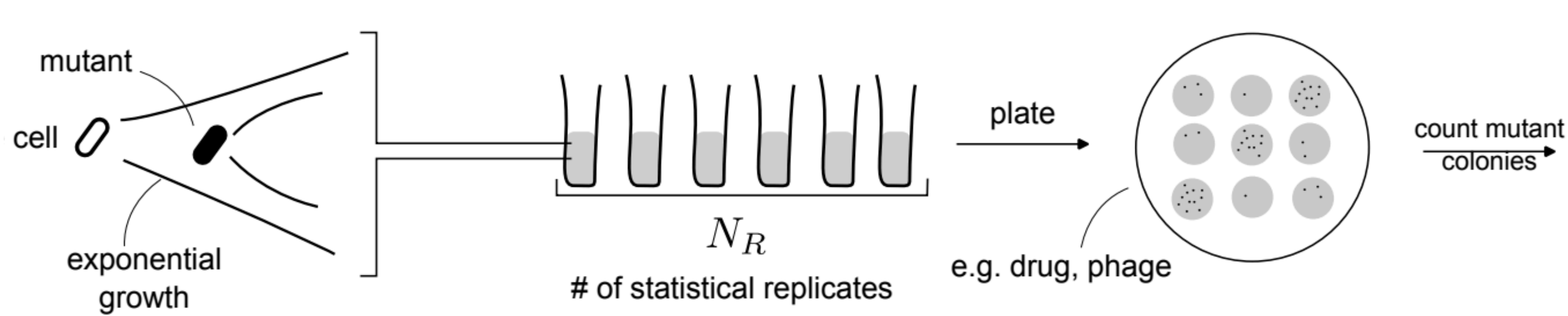
Numerical simulation result



data
(1943)

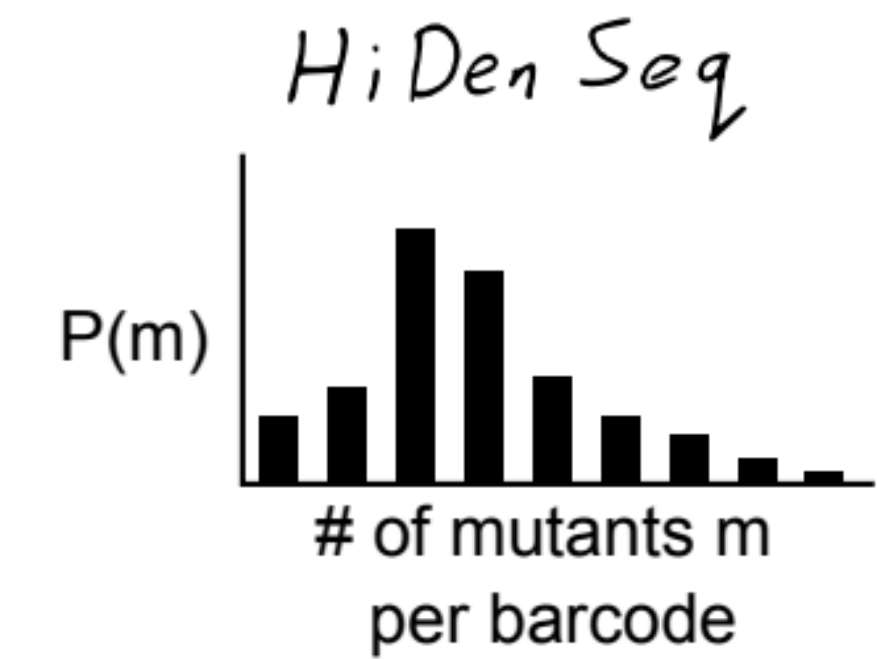
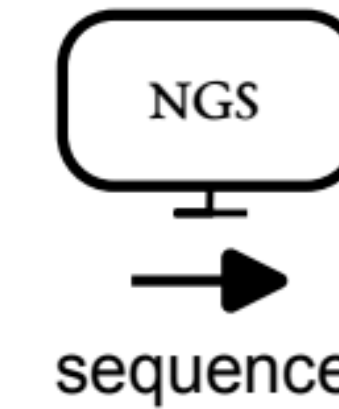
EXPERIMENT NO.	I	IO	II	I5	I6	I7	2Ia	2Ib
Number of cultures	9	8	IO	IO	20	12	19	5
Volume of cultures, cc	10.0	10.0	10.0	10.0	.2*	.2*	.2	10.0
Volume of samples, cc	.05	.05	.05	.05	.08	.08	.05	.05
<i>Culture No.</i>								
I	IO	29	30	6	I	I	0	38
2	18	41	10	5	0	0	0	28
3	125	17	40	10	3	0	0	35
4	10	20	45	8	0	7	0	107
5	14	31	183	24	0	0	8	13
6	27	30	12	13	5	303	I	
7	3	7	173	165	0	0	0	
8	17	17	23	15	5	0	I	
9	17		57	6	0	3	0	
10			51	10	6	48	15	
11					107	I	0	
12					0	4	0	
13					0		19	
14					0		0	
15					I		0	
16					0		17	
17					0		11	
18					64		0	
19					0		0	
20					35			
Average per sample	26.8	23.8	62	26.2	11.35	30	3.8	48.2
Variance (corrected for sampling)	1217	84	3498	2178	694	6620	40.8	1171

experiment (2026)

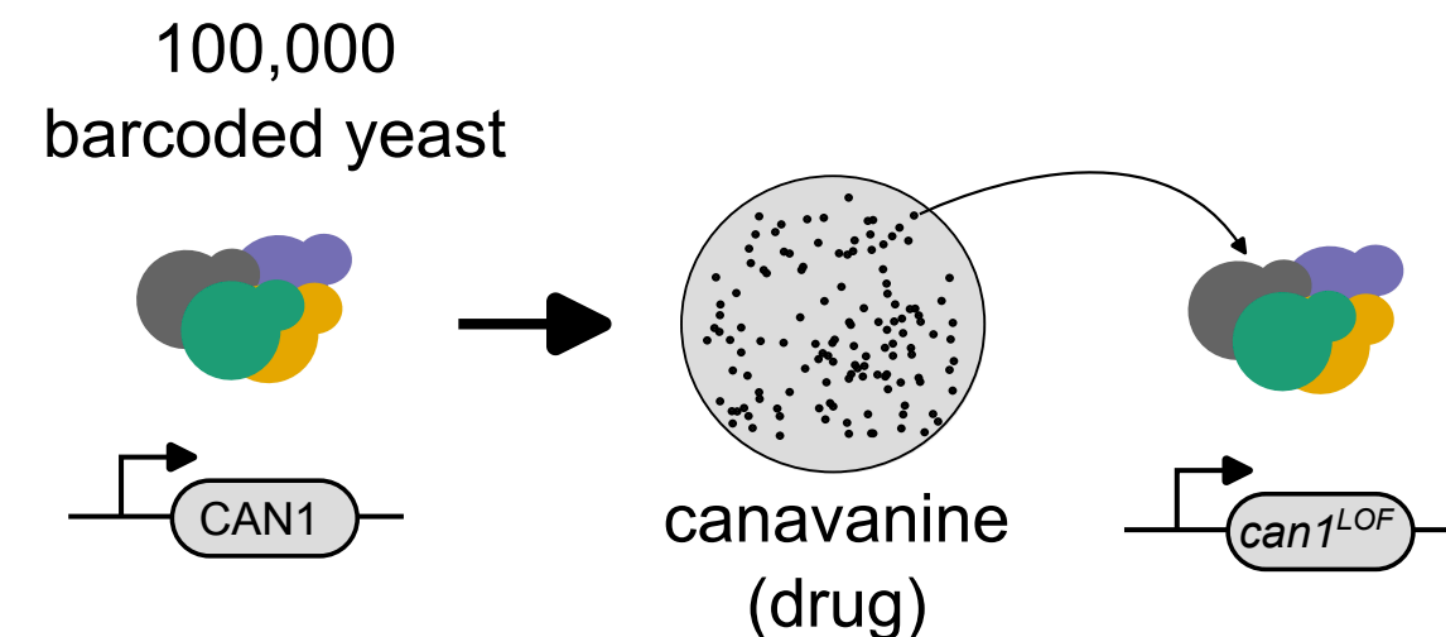


select
for mutants

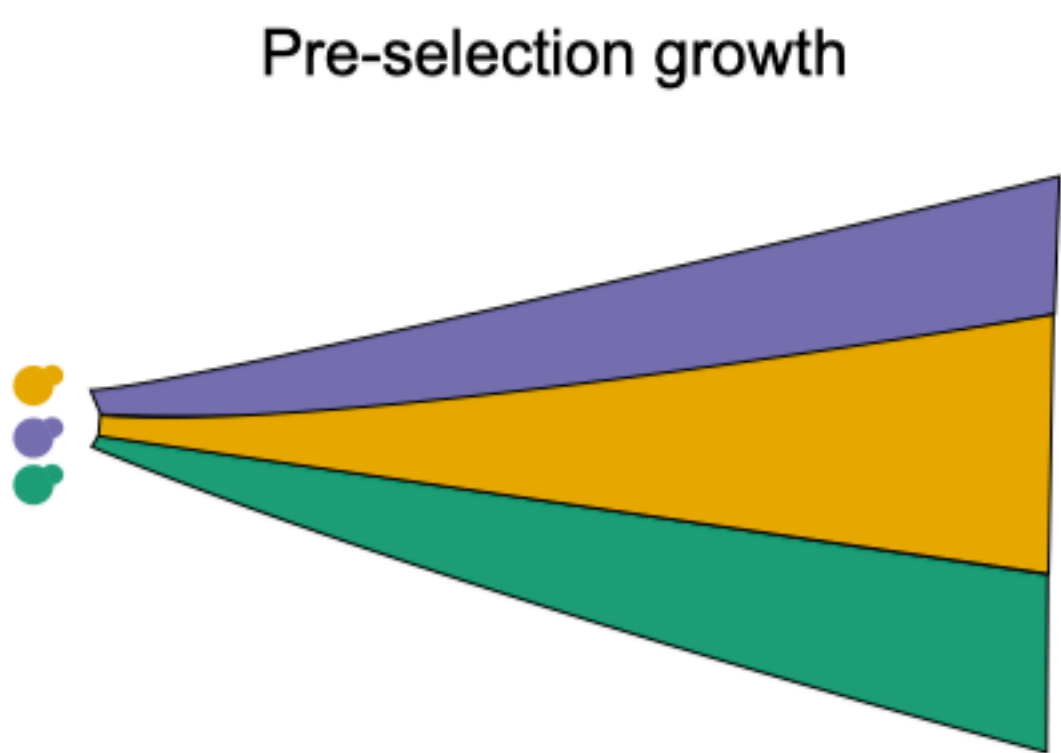
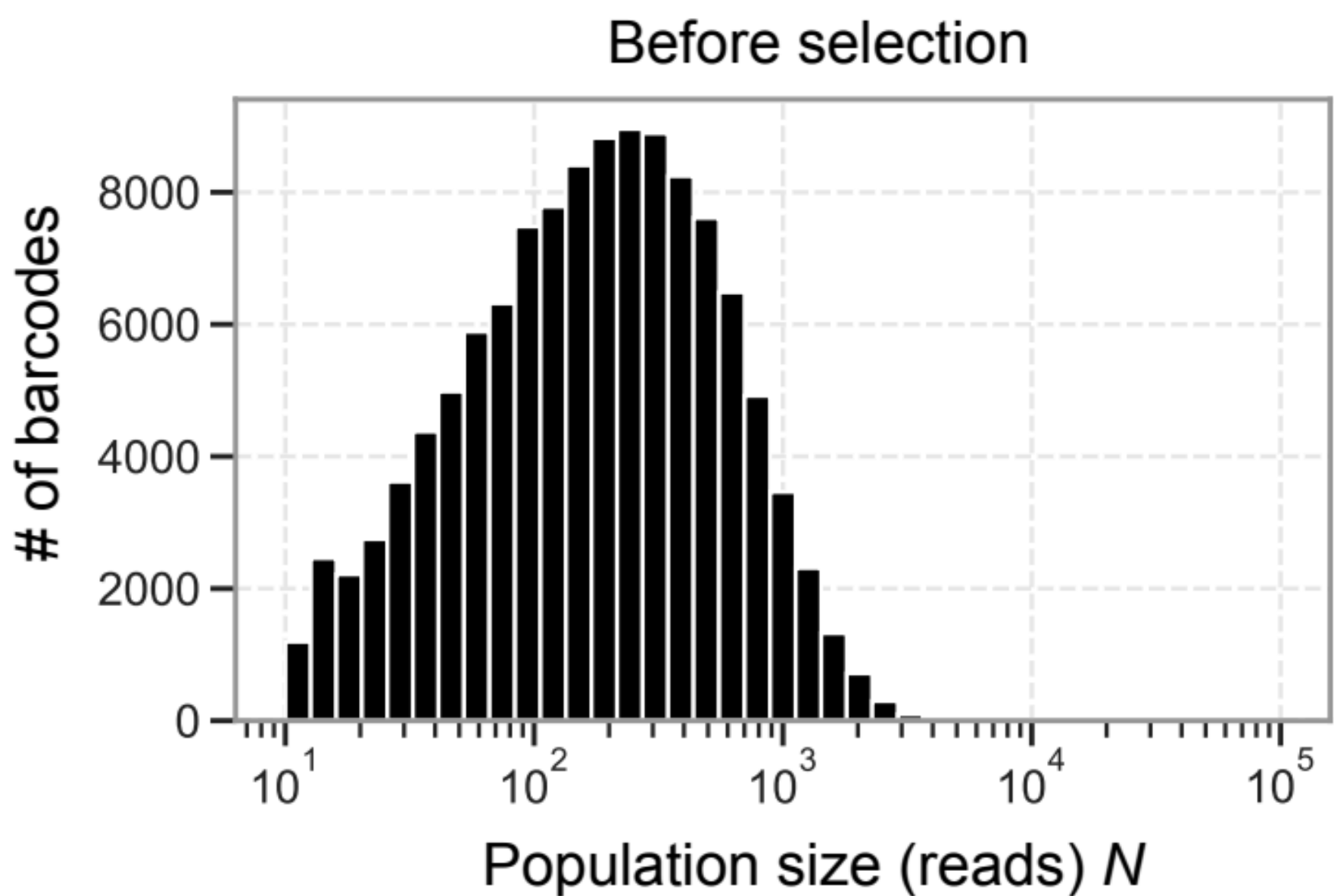
e.g. drug, phage



compare: Luria & Delbruck, 1943: $N_R \sim 100$.
robots, 2010s: $N_R \sim 1000$



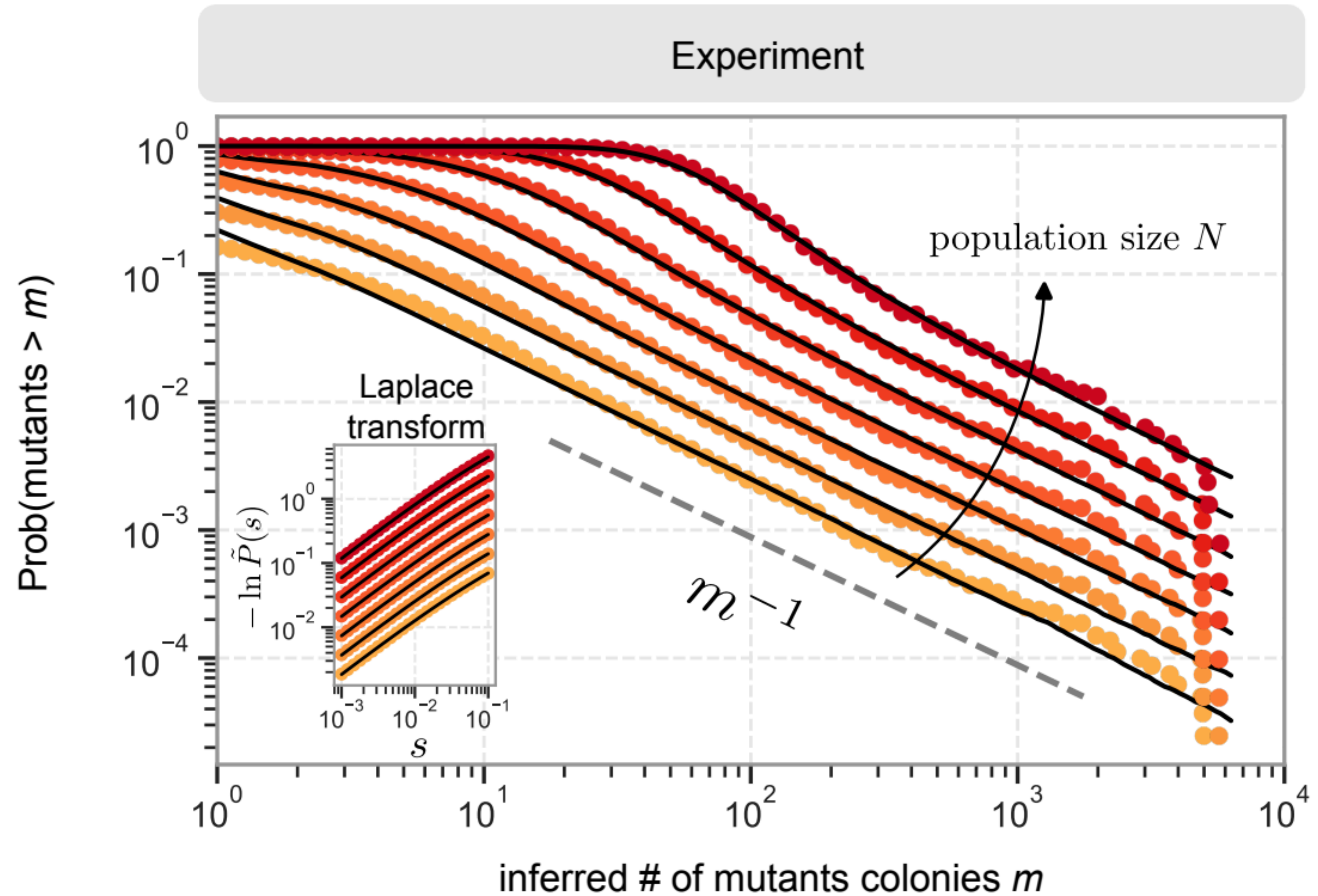
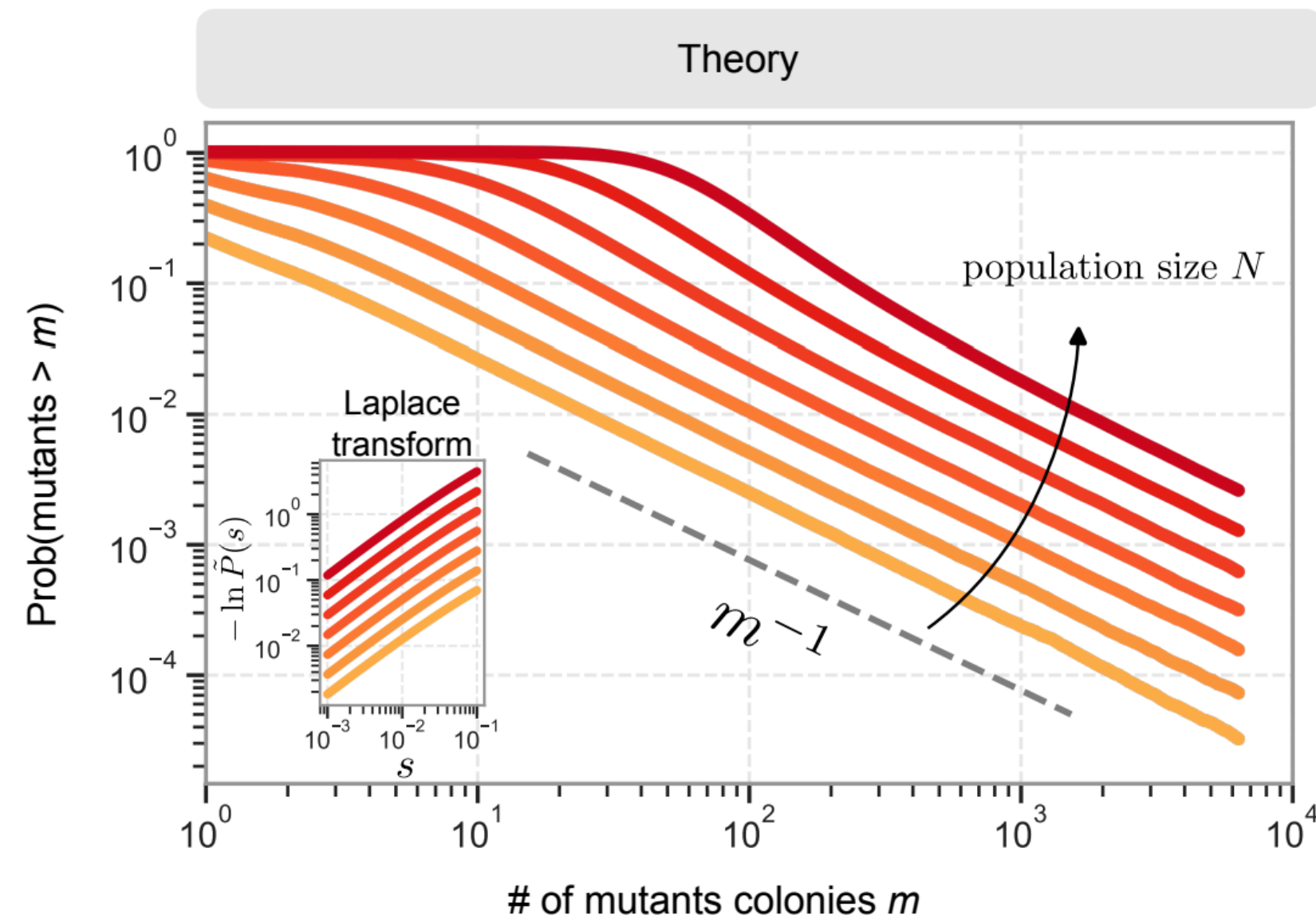
Barcodes to virtual lineages



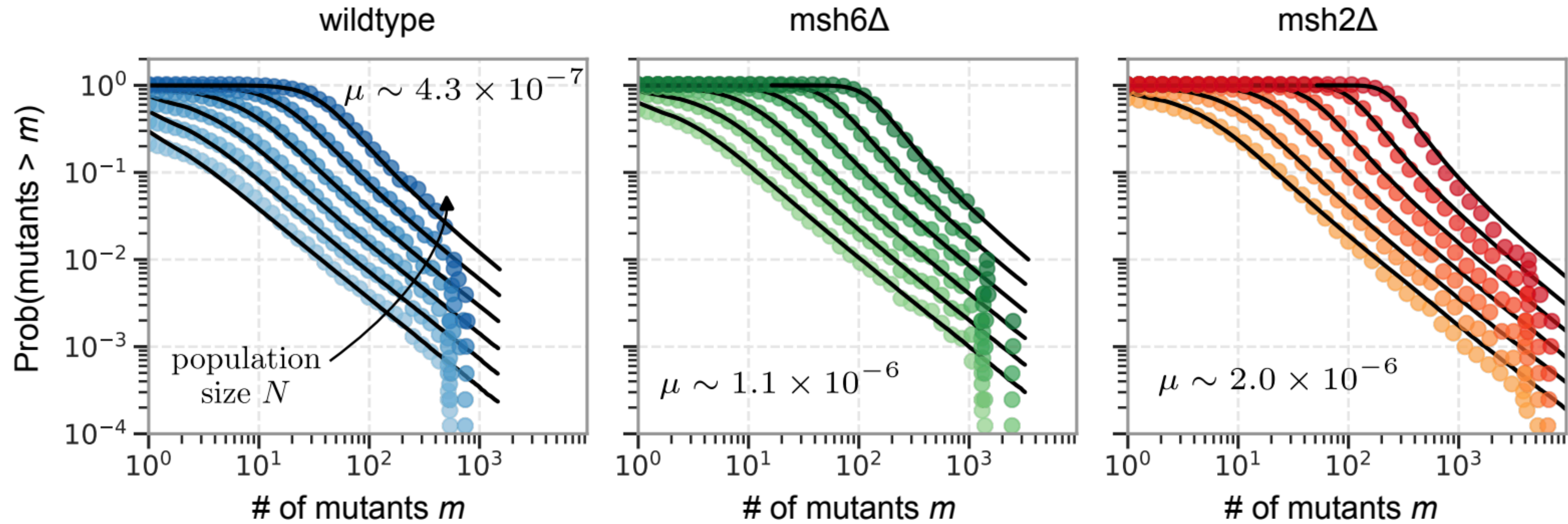
→ $n = 8$
barcoded
lineages

<i>barcode</i>	<i>reads</i>		<i>reads per virtual lineage</i>		
ATAGAC	52	————→ combine	52 + 102 = 154	<i>k</i> = 4 virtual lineages	
GGACAT	102				
TCCATG	31	————→	31 + 38 + 83 = 152		
CGTCAG	38				
GACGAT	83				
AATGCG	152	————→	152		
GGCAAC	65	————→	65 + 90 = 165		
GCATTA	90				

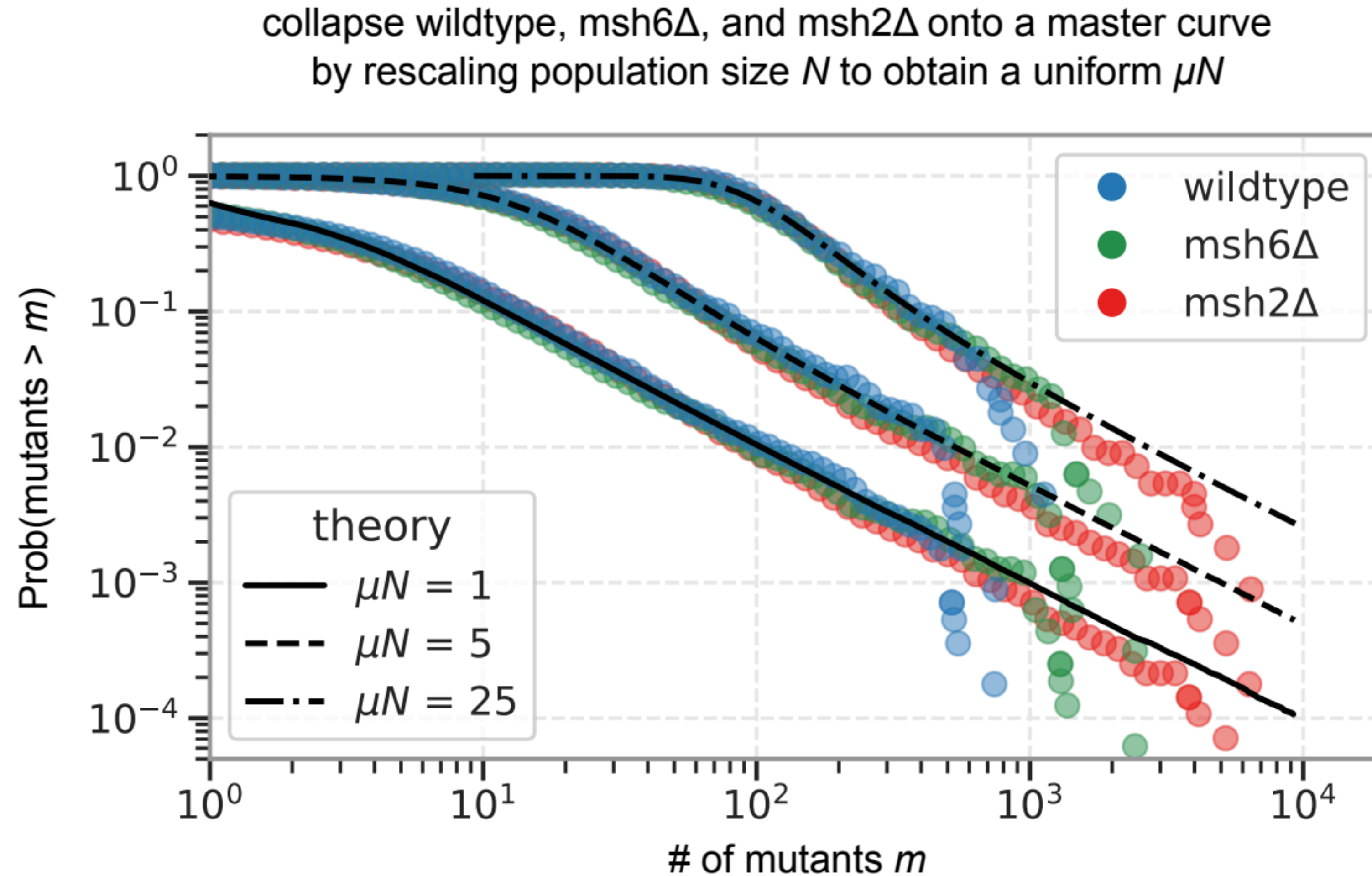
LD distribution



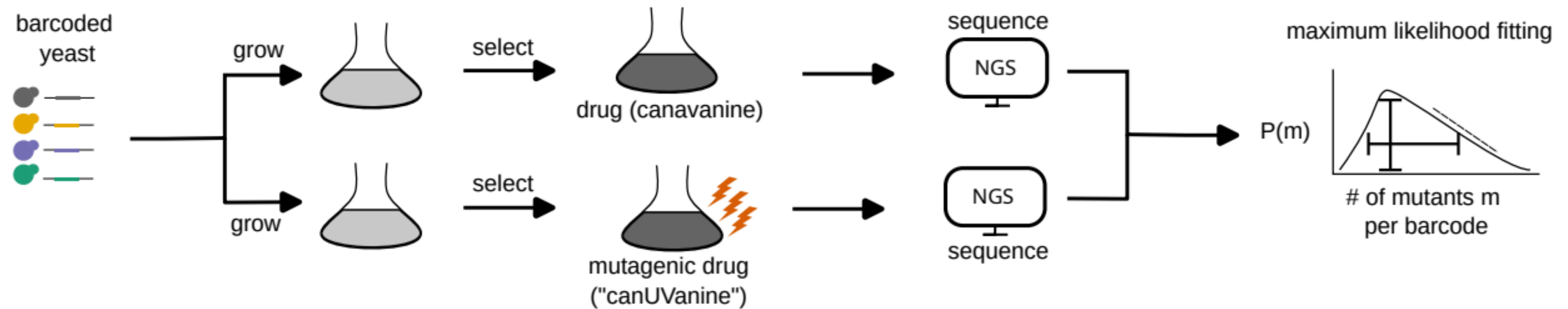
DNA mismatch repair mutants have different distribution



Distributions collapse as predicted



Mutagenic drug destroys pattern



c

Fit: spontaneous mutagenesis only

